

Six-hour M5 study — final snapshot COMPLETE

Snapshot generated 2026-10-07T21:36:28+00:00 (UTC). This report uses current results under this checkout; repository examples and dissertation narrative are not treated as outputs of this execution.

Overall execution status: COMPLETE. There are 372 completed current-scope configured runs and 30 recorded LLM calls in those completed runs. A completed deterministic control is not an LLM agent experiment. This is the latest available report; a PARTIAL or BLOCKED report is not a completed experiment.

Business results at this snapshot

Completed numerical evidence: six_hour_numeric_30seeds, 360 runs, 30 simulation seeds, seven decision days and all thirty selected series. These are simulated operational tradeoffs, not measured retailer outcomes.

Policy / scenario	Mean cost	Mean fill	Hard violations	Holds	Reference deviations
B1 / derived_field_collapse	621.067	0.928	0	0	179
B1 / feed_gap	443.184	0.908	0	0	210
B1 / normal	399.847	0.920	0	0	209
B2 / derived_field_collapse	548.971	0.852	0	0	122
B2 / feed_gap	440.066	0.772	0	0	131
B2 / normal	399.629	0.810	0	0	161
B3 / derived_field_collapse	497.046	0.937	0	0	78
B3 / feed_gap	425.570	0.903	63	0	37
B3 / normal	375.362	0.940	0	0	10
B4 / derived_field_collapse	369.423	0.905	0	90	6
B4 / feed_gap	367.796	0.905	0	90	5
B4 / normal	374.000	0.939	0	0	11

In the feed-gap simulation, B3 recorded 63 actual hard violations and 0 holds; the gated B4 recorded 0 hard violations and 90 holds. Mean fill was 0.903 for B3 and 0.905 for B4. The gate can reduce execution while changing service and cost. These exploratory observations do not prove long-run superiority or adequate statistical power.

Actual local-model feasibility evidence: ega-qwen2.5:1.5b-sixhour; 4 completed B9/B10 runs, 30 recorded calls, 0 executed decisions and 8 held decisions. The completed cases contain invalid or incomplete grounding, including recorded dimensional mismatches. Fail-closed validation prevents those invalid extractions from proceeding. An executed LLM replenishment benefit has not been established by these held cases. These findings concern this small local model, serving profile and controlled corpus; they do not generalize to all LLM systems.

Strict post-hoc grounding measured 88 exact eleven-field tuples among 95 returned tuples and 112 actually submitted source rules: precision=0.926, recall=0.786. It identified 7 wrong tuples and 24 omitted submitted tuples; the recorded field mismatches were unit errors. Only 16 of 157 available documents were submitted on each of 7 extraction days before early abort; later unprocessed documents are coverage gaps. One fault-gated day had no extraction. These are descriptive repeated exposures, not independent grounding trials.

Initial inventory can satisfy two days of sales even when every replenishment decision is held. Purchase expenditure is charged immediately, terminal stock has no salvage credit, and shortage/replenishment effects beyond the window are omitted. Structured controls can therefore buy inventory for future days while all-held prose cases appear cheaper. Low short-window cost must not be called LLM savings or evidence of equivalent long-run service. Mean inventory, holds and execution are reported alongside cost/fill. The seven-day numerical template grid, two-day prose parser comparators, live model arms and separate two-day structured-rule controls retain their own horizons and carriers.

Two-day stage / policy / scenario	Cost	Fill	Mean inventory	Exec / held
six_hour_agentic_prose / B10 / derived_field_collapse	8.513	1.000	237.500	0 / 2
six_hour_agentic_prose / B10 / normal	8.513	1.000	237.500	0 / 2
six_hour_agentic_prose / B3 / derived_field_collapse	8.513	1.000	237.500	0 / 2
six_hour_agentic_prose / B3 / normal	8.513	1.000	237.500	0 / 2
six_hour_agentic_prose / B4 / derived_field_collapse	8.513	1.000	237.500	0 / 2
six_hour_agentic_prose / B4 / normal	8.513	1.000	237.500	0 / 2
six_hour_agentic_prose / B9 / derived_field_collapse	8.513	1.000	237.500	0 / 2
six_hour_agentic_prose / B9 / normal	8.513	1.000	237.500	0 / 2
six_hour_structured_controls / B3 / derived_field_collapse	178.272	1.000	231.000	2 / 0
six_hour_structured_controls / B3 / normal	97.066	1.000	224.000	2 / 0
six_hour_structured_controls / B4 / derived_field_collapse	67.673	1.000	231.000	1 / 1
six_hour_structured_controls / B4 / normal	96.499	1.000	225.000	2 / 0

Managerial interpretation is conditional: grounding reliability, actual hard violations, holds, service and latency must be assessed together. Historical M5 sales from thirty selected FOODS item-store series are combined with simulated inventory, suppliers and replenishment; no organizational ROI, field deployment or human-review benefit is measured.

Historical study dates are distinct from cloud execution timestamps: the registered model-training data end on 2016-01-18 (index1815, M5d_1816); numerical decisions cover 2016-02-02 through 2016-02-08 (indices1830-1836, d_1831-d_1837); two-day model/control decisions cover 2016-02-02 through 2016-02-03. The prepared panel's calendar.csv supplies the per-stage mappings below; 2026 timestamps record execution and report preparation.

Grounding safeguard limit relevant to RQ3/H3: the deployed prose verifier checks source references, entity allowlists, dimensions/ranges, conflicts and required coverage. Full source-value/scope/precedence equality is checked only for the deterministic RULE carrier. Holding the observed unit errors does not prove that every semantically wrong prose field would be caught. The strict inverse-prose assessor is an after-the-fact measurement, not a deployed repair or a hidden deterministic replacement for the model. No model-grounded proposal reached the optimizer in these held live cases.

Reduced protocol and six-hour scope

Document status: final bounded-study snapshot. Execution and missing-data labels remain factual; finalization of this document does not turn unfinished experiments into completed runs or complete the full revised dissertation.

The user changed the objective to a study bounded to six hours while preserving all thirty prepared M5 item-store series. The long-stage studies are retained as auxiliary history and are not pooled into this study's run count, policy estimates or significance claims. This shortened protocol does not complete the full revised dissertation, thirty-seed confirmatory comparisons, the complete rolling-origin/severity matrix, or the human-audit study.

This report describes the bounded numerical controls, actual live-model grounding checks, and closed-loop LLM decisions completed within the registered scope. Planned or deadline-interrupted runs remain unrun/partial. There are no paired significance or thirty-seed power claims for this shortened study. Short decision horizons can hide service effects through warmup stock and lead times, so small cost/fill differences must not be generalized to long-run replenishment.

requested at: 2026-10-07T20:01:00+00:00 (UTC).

experiment cutoff: 2026-10-08T01:31:00+00:00 (UTC).

report deadline: 2026-10-08T02:01:00+00:00 (UTC).

Bounded protocol limit: Shorter study cannot establish the full dissertation hypotheses or replace thirty-seed three-origin long-run evidence

Bounded protocol limit: All numerical and eventual live agentic arms use the same reduced solver settings; no cross-configuration pooling

Bounded protocol limit: Final report records actual completed grids and marks any unfinished cells at the hard deadline

Frozen numerical subset: policies=['B1', 'B2', 'B3', 'B4']; scenarios=['normal', 'derived_field_collapse', 'feed_gap']; seeds=8; origins=1; decision days=7; configured runs=96. All thirty prepared series are retained. Forecast/solver settings are read from each actual run config and are not pooled with the old long-study settings.

Frozen live-model subset: {"carrier": "controlled_prose", "config": "results/run_configs/six_hour_agentic_prose.yaml", "days": 2, "document_batch_size": 4, "expected_decisions": 16, "expected_runs": 8, "origins": 1, "policies": ["B3", "B4", "B9", "B10"], "profile": "original repository FORMAT_CONVENTIONS; no worker sidecar", "scenarios": ["normal", "derived_field_collapse"], "seeds": [0], "source_document_count_per_normal_day": 157}. It uses the original repository FORMAT_CONVENTIONS and the recorded common model digest. The separately tested documented-format variant was not selected for evaluation.

Development selection disclosure: {"disclaimer": "Development selection is disclosed; this small sample does not validate general grounding accuracy", "documented_format_variant": "0/4 exact for each carrier; retained as development only", "qwen15_base_batch4_prose_exact_rules": "3/4; unit_cost unit error", "qwen15_base_template_exact_rules": "0/4; precedence error", "selection": "Base batch4 for practical throughput and stronger observed prose grounding; model/profile frozen before evaluation", "window": "day1700, simulationseed4242; no evaluation labels sent to model"}

Fault timing: Derived-field collapse begins at start_day+1; two-day LLM study includes a clean first day and an active fault on the second. Seven-day numeric study covers the longer episode.

Separate structured-rule short-window control: {"added_at_utc": "2026-10-07T20:48:52.635288+00:00", "carrier": "template", "config": "results/run_configs/six_hour_structured_controls.yaml", "days": 2, "expected_decisions": 8, "expected_runs": 4, "origins": 1, "policies": ["B3", "B4"], "purpose": "Exact same information with deterministic structured-rule extraction; distinguish replenishment behavior from failure to parse controlled prose", "scenarios": ["normal", "derived_field_collapse"], "seeds": [0], "selection_disclosure": "Added after the first four prose controls demonstrated the documented deterministic parser limitation, before observing completed LLM outcomes. No model/prompt/settings change."}

Numerical extension registration: {"analytical_view": "results/six_hour_numeric_30seeds", "concurrency_note": "Two numerical workers selected before extension launch after all local LLM runs completed; one CPU thread each, no concurrent inference. Dataset, seed grid, solver and forecast remain unchanged.", "config": "results/run_configs/six_hour_numeric_extension.yaml", "config_sha256": "d400722b69a70caab08615840a5bea23019e3f9784dc243ef5c777897ceb2e88", "days": 7, "expected_decisions": 1848, "expected_runs": 264, "frozen_at_utc": "2026-10-07T20:58:21.469540+00:00", "origins": 1, "owner_helper_sha256": "33dd70035ede229cb48f1785e12a90f47ac9dff4747ff4afd09cb104b20bb7b1", "owner_validation_checks": 25, "pooling_requirements": "No settings difference except output and seeds; verify exact disjoint360cells, actual training tensors and LightGBM numerical trees, source/data/config hashes. Preserve original source paths and manifests. No LLM30seed claim.", "reason": "Base96 actual runtime20minutes leaves sufficient deadline budget for a disjoint seed extension; all four policies and all three scenarios retained. This is an exploratory post-launch replication increase, not a confirmatory dissertation registration. Base results had already completed when extension was selected.", "reporting": "Source phases reported separately until fully complete and equivalence proven; count source identities once in verified analytical union. No formal efficacy/power/significance claim.", "seeds": [8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29], "workers": 2}. The

completed initial cohort was available before extension launch. Expansion was motivated by measured runtime capacity, and the combined analysis remains exploratory rather than a confirmatory dissertation registration.

Execution status and remaining work

COMPLETE means every configured policy/scenario/seed/origin run has a per-run summary and the configured number of daily rows. PARTIAL means execution has started or some runs are complete. BLOCKED records a stated missing prerequisite. UNRUN means no completed run or startup evidence is available. Completion counts are calculated from artifacts, even when pipeline status is stale; completion does not by itself establish audit validity, statistical power, or model quality. Overall execution completion applies to stages marked required; superseded or archived studies retain their actual artifacts separately. Configured-run completion also does not satisfy unimplemented dissertation requirements.

Stage	Status	Required	Complete / target	Decision days	Policies	Carrier
m5_smoke	COMPLETE	no	12 / 12	4	B1, D0, D1	template
ml_m5_smoke	COMPLETE	no	16 / 16	4	B1, B2, B3, B4	template
baseline_pilot	COMPLETE	no	16 / 16	7	B1, B2, B3, B4	template
main_study_stage1	PARTIAL	no	20 / 480	28	B1, B2, B3, B4	template
foundation_30series	BLOCKED	no	0 / 360	28	B1, B5	template
llm_architectures_ollama	BLOCKED	no	0 / 18	4	B4, B6, B7, B8, B9, B10	template
llm_prose_reference_30series	PARTIAL	no	1 / 360	28	B3, B4	template
llm_prose_study_ollama	BLOCKED	no	0 / 540	28	B4, B9, B10	prose
llm_prose_study_reference	COMPLETE	no	120 / 120	4	B3, B4	template
llm_template_architectures_30series	BLOCKED	no	0 / 900	28	B6, B7, B8, B9, B10	template
parallel_runner_validation	COMPLETE	no	8 / 8	1	B1, B2, B3, B4	template
six_hour_agentic_prose	COMPLETE	yes	8 / 8	2	B3, B4, B9, B10	prose
six_hour_numeric	COMPLETE	yes	96 / 96	7	B1, B2, B3, B4	template
six_hour_numeric_30seeds	COMPLETE	no	360 / 360	7	B1, B2, B3, B4	template
six_hour_numeric_extension	COMPLETE	yes	264 / 264	7	B1, B2, B3, B4	template
six_hour_structured_controls	COMPLETE	yes	4 / 4	2	B3, B4	template

main_study_stage1: pipeline-reported status=paused; observed artifact status=PARTIAL. In-progress run directories without complete summaries: 3.

foundation_30series Auxiliary stage's historically reported prerequisite: Chronos metadata pinned and libraries validated; remaining weight file requires us.aws.cdn.hf.co network route

foundation_30series readiness observation: Pinned Chronos metadata revision has been obtained; it does not attest downloaded model weights or tested foundation inference.

foundation_30series: pipeline-reported status=blocked; observed artifact status=BLOCKED. In-progress run directories without complete summaries: 0.

llm_architectures_ollama Auxiliary stage's historically reported prerequisite: Official model registered; awaiting live development grounding and throughput/storage preflight

llm_architectures_ollama: pipeline-reported status=blocked; observed artifact status=BLOCKED. In-progress run directories without complete summaries: 0.

llm_prose_reference_30series: pipeline-reported status=paused; observed artifact status=PARTIAL. In-progress run directories without complete summaries: 1.

llm_prose_study_ollama Auxiliary stage's historically reported prerequisite: Official model registered; awaiting live development grounding and throughput/storage preflight

llm_prose_study_ollama: pipeline-reported status=blocked; observed artifact status=BLOCKED. In-progress run directories without complete summaries: 0.

llm_prose_study_reference: pipeline-reported status=complete; observed artifact status=COMPLETE. In-progress run directories without complete summaries: 0.

llm_template_architectures_30series Auxiliary stage's historically reported prerequisite: Official model registered; awaiting live development grounding and throughput/storage preflight

llm_template_architectures_30series: pipeline-reported status=planned; observed artifact status=BLOCKED. In-progress run directories without complete summaries: 0.

parallel_runner_validation: pipeline-reported status=complete; observed artifact status=COMPLETE. In-progress run directories without complete summaries: 0.

six_hour_agentic_prose readiness observation: Verified registry download and local registration are attested by ollama_model_identity.json; this alone does not validate functional inference or full-study readiness.

six_hour_agentic_prose readiness observation: Configured model digest matches registered identity.

six_hour_agentic_prose: pipeline-reported status=complete; observed artifact status=COMPLETE. In-progress run directories without complete summaries: 0.

six_hour_numeric retains its original phase provenance and is represented in the verified six_hour_numeric_30seeds analytical view. Its runs are counted once in report totals and policy summaries.

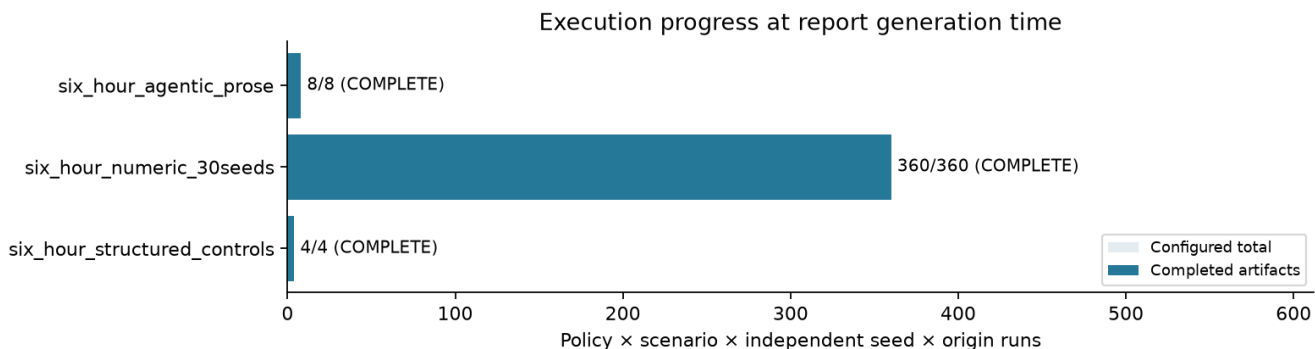
six_hour_numeric: pipeline-reported status=complete; observed artifact status=COMPLETE. In-progress run directories without complete summaries: 0.

six_hour_numeric_30seeds is a verified external-source analytical union, not another experiment. It covers the disjoint original cohorts only after complete source coverage, matching immutable configs/data/environment and exact actual trained states across every merged and worker model. Original run paths and hash-verified archives remain unchanged.

six_hour_numeric_extension retains its original phase provenance and is represented in the verified six_hour_numeric_30seeds analytical view. Its runs are counted once in report totals and policy summaries.

six_hour_numeric_extension: pipeline-reported status=complete; observed artifact status=COMPLETE. In-progress run directories without complete summaries: 0.

six_hour_structured_controls: pipeline-reported status=complete; observed artifact status=COMPLETE. In-progress run directories without complete summaries: 0.



Configured run completion; worker and merged copies of the same run are counted once.

Model preparation and functional readiness

Recorded local model: ega-qwen2.5:1.5b-sixhour; registered digest: 14bdd46019879e18cd27f21b46e2c377bf801e827e2abb46be2daedcedaf0f64; parameter size: 1.5B; quantization: Q4_K_M. Source attestation: unrecorded. This establishes the recorded verified download/registration step, not successful end-to-end experimental decisions.

Readiness records below are separate from completed simulation results: model registration, validated structured responses, measured throughput, configuration freeze and a completed study are different milestones. Historical network failures in earlier audit/validation snapshots do not override a later verified identity. Successful preflight calls do not count as completed full-study runs.

retained auxiliary preparation source:
results/local_model_development_30series_batch1/development_manifest.json

Readiness summary: {"complete": true, "day": 1700, "development_only": true, "model_revision": "bd9b617abc6c2ec05ebeac915201900320377281168b3949032799cc8f20d51d", "pilot_runs": 0, "series": 30}

retained auxiliary preparation source: results/logs/ollama_cloud30_functional_readiness.json

Readiness summary: {"elapsed_seconds": 1.1468176409998705, "model_revision": "bd9b617abc6c2ec05ebeac915201900320377281168b3949032799cc8f20d51d"}

Recorded functional check: requested structured value matched=True; positive actual input/output usage=True; prompt tokens=36; completion tokens=8; elapsed seconds=1.147. This validates this small structured call, not supplier-rule accuracy, inventory performance or full-study throughput.

current bounded-study preparation source: results/logs/six_hour_preflight.json

This readiness snapshot records agentic configs awaiting freeze/registration:
/workspace/MasterDissertation/results/run_configs/six_hour_agentic_prose.yaml,
/workspace/MasterDissertation/results/run_configs/six_hour_agentic_template.yaml. Registered stage configs and actual run artifacts take precedence if this preparation snapshot later becomes stale.

retained auxiliary preparation source: results/run_configs/local_model_resource_preflight.json

Readiness summary: {"model_revision": "bd9b617abc6c2ec05ebeac915201900320377281168b3949032799cc8f20d51d"}

Measured resource preflight: passed=False; blockers=['Projected artifact storage plus reserve exceeds current free disk']. Measurements: {"max_actual_zip_deflate_sample_ratio": 0.15718854449707842, "mean_call_seconds": 92.26134998425005, "mean_request_artifact_bytes": 13287.5, "p95_call_seconds": 123.14005513200027, "real_call_records": 4}. Successful-path model-only projection at mean=6,006.214 days, at p95=8,016.418 days. These are development-sample planning estimates, not elapsed full-experiment results; they exclude additional training, optimization, scheduling, retries and reports.

retained auxiliary preparation source: results/run_configs/local_model_resource_preflight_incremental.json

Incremental resource evidence: 2 actual development responses; first extraction request=243.206 seconds; planned successful-path requests before retries=438,480. The recorded same-latency multiplication is 1,234.271 days. This is a scale illustration from an initial request, not a representative completion forecast or an executed study duration.

Incremental estimate limit: A single extraction request is not a representative latency estimate for all roles, memory lengths, cache effects, validation failures and retries.

Incremental estimate limit: The coarse same-latency multiplication is a scale illustration, not a completion forecast.

Incremental estimate limit: Full inference cannot be called feasible based on an installed model or a trivial response.

Incremental estimate limit: No full-study LLM evaluation has started; CPU/GPU/provider choice and common serving batch remain development decisions.

current bounded-study preparation source: results/run_configs/six_hour_runtime_preflight.json

This readiness snapshot records agentic configs awaiting freeze/registration:
/workspace/MasterDissertation/results/run_configs/six_hour_agentic_template.yaml. Registered stage configs and actual run artifacts take precedence if this preparation snapshot later becomes stale.

current bounded-study preparation source: results/run_configs/six_hour_supervisor_status.json

Readiness summary: {"phase": "all_supplied_studies_complete", "updated_at_utc": "2026-10-07T20:56:49.458868+00:00"}

This readiness snapshot records agentic configs awaiting freeze/registration:
/workspace/MasterDissertation/results/run_configs/six_hour_agentic_template.yaml. Registered stage configs and actual run artifacts take precedence if this preparation snapshot later becomes stale.

retained auxiliary preparation source: results/run_configs/supervisor_status.json

Readiness summary: {"blocker": "Projected artifact storage plus reserve exceeds current free disk; detailed truthful estimates saved in local_model_resource_preflight.json", "functional_usage": {"completion_tokens": 8, "prompt_tokens": 36, "prompt_tokens_details": {"cached_tokens": 35}, "total_tokens": 44}, "model_name": "ega-llama3.2:3b-cloud30", "model_revision": "bd9b617abc6c2ec05ebeac915201900320377281168b3949032799cc8f20d51d", "phase": "blocked", "preflight_passed": false, "updated_at_utc": "2026-10-07T20:07:08.033226+00:00"}

Recorded foundation metadata: amazon/chronos-t5-tiny at revision 29d808298f1a62493e7b9a5e08529d0d930fa189. Metadata retrieval is not evidence that the model.safetensors download or inference succeeded.

Controlled grounding benchmarks — actual current artifacts

Incremental and final snapshots from the same development folder can describe the same responses. They are shown as recorded snapshots and are not pooled or counted as independent replications. Retained development configurations with different batch sizes also remain separate from frozen evaluation studies.

Source: results/grounding_templates/grounding_results.json; evidence role=retained auxiliary grounding; reader mode=deterministic_template; model=not recorded; model revision=not recorded; development_only=False; incremental=False; complete=not recorded; total LLM calls=0; errors=0; tokens=0. Controlled carrier benchmark, not a validated natural-language dataset. No expected labels are sent to the model.

Case / carrier	Whole set exact	Tuple precision	Tuple recall	False / omitted	Errors eligible for solver	Escalated
normal / deterministic_template	True	1.000	1.000	0 / 0	0	False
aggregate_moq / deterministic_template	True	1.000	1.000	0 / 0	0	False
foreign_unit_moq / deterministic_template	True	1.000	1.000	0 / 0	0	False
pack_change / deterministic_template	True	1.000	1.000	0 / 0	0	False

Case / carrier	Whole set exact	Tuple precision	Tuple recall	False / omitted	Errors eligible for solver	Escalated
capacity_cut / deterministic_template	True	1.000	1.000	0 / 0	0	False
eligibility_change / deterministic_template	True	1.000	1.000	0 / 0	0	False

Source: results/local_model_development_30series_batch1/grounding/grounding_results.json; evidence role=retained auxiliary grounding; reader mode=local LLM development sample; model=not recorded; model revision=bd9b617abc6c2ec05ebeac915201900320377281168b3949032799cc8f20d51d; development_only=True; incremental=False; complete=not recorded; total LLM calls=4; errors=0; tokens=11690. Field-complete controlled carriers; this sample is not a natural-language production benchmark.

Case / carrier	Whole set exact	Tuple precision	Tuple recall	False / omitted	Errors eligible for solver	Escalated
development batch / template	False	—	—	1 / 1	—	—
development batch / template	False	—	—	1 / 1	—	—
development batch / prose	False	—	—	1 / 1	—	—
development batch / prose	False	—	—	1 / 1	—	—

Source: results/local_model_development_30series_batch16/grounding/grounding_results.json; evidence role=retained auxiliary grounding; reader mode=local LLM development sample; model=not recorded; model revision=bd9b617abc6c2ec05ebeac915201900320377281168b3949032799cc8f20d51d; development_only=True; incremental=False; complete=not recorded; total LLM calls=2; errors=0; tokens=9659. Field-complete controlled carriers; this sample is not a natural-language production benchmark.

Case / carrier	Whole set exact	Tuple precision	Tuple recall	False / omitted	Errors eligible for solver	Escalated
development batch / template	False	—	—	0 / 14	—	—
development batch / prose	False	—	—	1 / 15	—	—

Source: results/local_model_development_30series_batch16/grounding/incremental_grounding_results.json; evidence role=retained auxiliary grounding; reader mode=local LLM development sample; model=not recorded; model revision=bd9b617abc6c2ec05ebeac915201900320377281168b3949032799cc8f20d51d; development_only=True; incremental=True; complete=False; total LLM calls=not recorded; errors=not recorded; tokens=not recorded. Labels are compared by evaluator only; they were not provided to the model. Schema validity does not establish grounding correctness.

Case / carrier	Whole set exact	Tuple precision	Tuple recall	False / omitted	Errors eligible for solver	Escalated
development batch / template	False	1.000	0.067	0 / 14	—	—
development batch / prose	False	0.000	0.000	1 / 15	—	—

Carrier	Source docs	Constraints returned	Schema valid / finish	Input / output tokens	Seconds
template	15	1	True / stop	4812 / 152	243.206

Carrier	Source docs	Constraints returned	Schema valid / finish	Input / output tokens	Seconds
prose	15	1	True / stop	4588 / 107	201.623

Schema-valid responses can still omit or misground rules. Missing residual-error/escalation fields remain unmeasured rather than zero. Development samples do not establish a favorable LLM effect or confirmatory full-study performance.

The measured development samples above contain actual extraction failures: whole-set mismatches, omissions or false constraints remain errors despite valid JSON and a normal finish reason.

Source: results/six_hour_development_grounding/grounding_results.json; evidence role=current bounded-study development; reader mode=real_llm; model=ega-qwen2.5:0.5b-sixhour; model revision=21539788ff12776c3460262e890e2d77f98d99e5ce24a1726a9582a31be23e44; development_only=True; incremental=False; complete=True; total LLM calls=4; errors=4; tokens=15686. Four controlled development cases with all 30 entities; not evaluation or general natural-language accuracy.

Case / carrier	Whole set exact	Tuple precision	Tuple recall	False / omitted	Errors eligible for solver	Escalated
development batch / template	False	—	—	— / —	—	—
development batch / template	False	—	—	— / —	—	—
development batch / prose	False	—	—	— / —	—	—
development batch / prose	False	—	—	— / —	—	—

Case / carrier / batch	Exact / false / omitted rules	Calls	Seconds	Request result
1 / template / 1	0 / unmeasured / 1	1	35.684	request failed
2 / template / 1	0 / unmeasured / 1	1	36.045	request failed
3 / prose / 1	0 / unmeasured / 1	1	32.902	request failed
4 / prose / 1	0 / unmeasured / 1	1	31.013	request failed

Development case 1 request error: LLM request failed after 1 attempts: Truncated model output; increase max_tokens or reduce batch size

Development case 2 request error: LLM request failed after 1 attempts: Truncated model output; increase max_tokens or reduce batch size

Development case 3 request error: LLM request failed after 1 attempts: Truncated model output; increase max_tokens or reduce batch size

Development case 4 request error: LLM request failed after 1 attempts: Truncated model output; increase max_tokens or reduce batch size

These exact-rule counts and durations come from the recorded development cases. Failed requests can have recorded omissions without a validated returned-rule set; missing false-rule/precision fields remain unmeasured. Serving-model selection uses these development samples, which are excluded from inventory-policy comparisons.

Source: results/six_hour_qwen15_development/grounding_results.json; evidence role=current bounded-study development; reader mode=real_llm; model=ega-qwen2.5:1.5b-sixhour; model revision=14bdd46019879e18cd27f21b46e2c377bf801e827e2abb46be2daedcedaf0f64; development_only=True; incremental=False; complete=True; total LLM calls=4; errors=0; tokens=14152. Four controlled development cases; serving batch selection only. Expected labels are excluded from all requests.

Case / carrier	Whole set exact	Tuple precision	Tuple recall	False / omitted	Errors eligible for solver	Escalated
development batch / template	False	—	—	— / —	—	—
development batch / prose	False	—	—	— / —	—	—
development batch / template	False	—	—	— / —	—	—
development batch / prose	False	—	—	— / —	—	—

Case / carrier / batch	Exact / false / omitted rules	Calls	Seconds	Request result
1 / template / 1	0 / 1 / 1	1	34.768	response received; see exact-rule counts
2 / prose / 1	0 / 1 / 1	1	30.906	response received; see exact-rule counts
3 / template / 4	0 / 4 / 4	1	55.303	response received; see exact-rule counts
4 / prose / 4	3 / 1 / 1	1	52.360	response received; see exact-rule counts

These exact-rule counts and durations come from the recorded development cases. Failed requests can have recorded omissions without a validated returned-rule set; missing false-rule/precision fields remain unmeasured. Serving-model selection uses these development samples, which are excluded from inventory-policy comparisons.

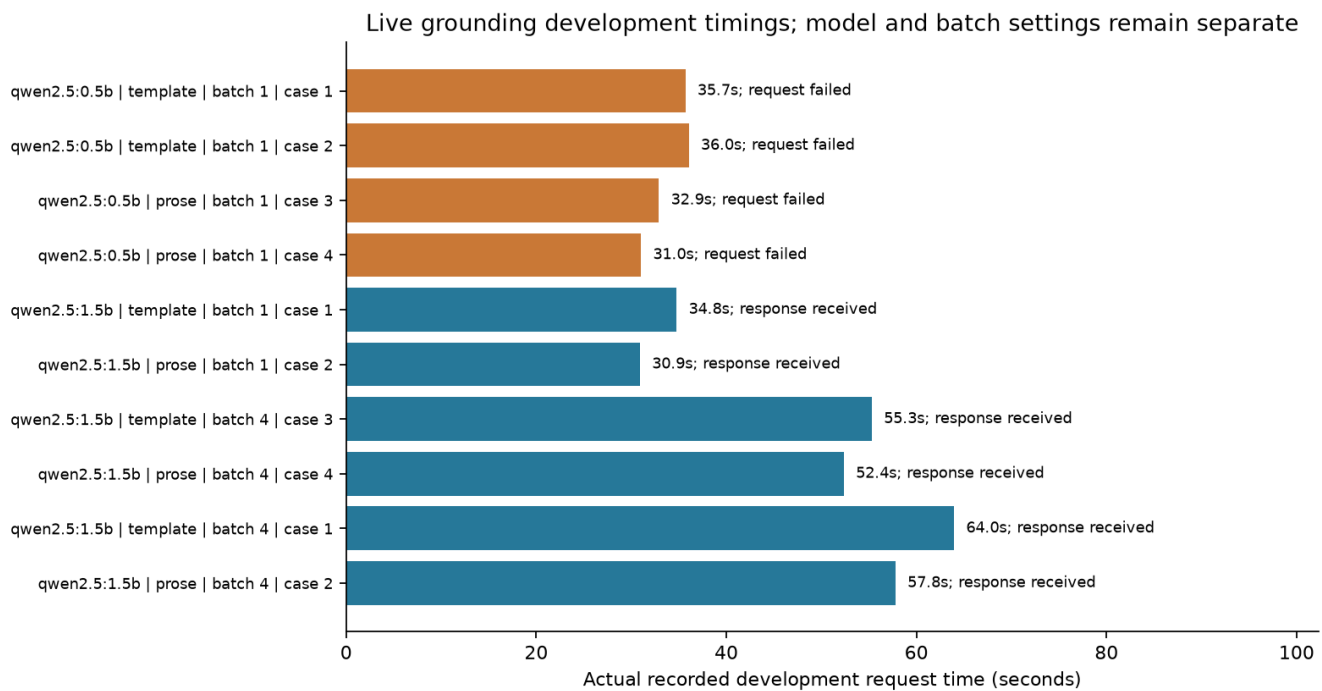
Source: results/six_hour_qwen15_documented_format_development/grounding_results.json; evidence role=current bounded-study development; reader mode=real_llm; model=ega-qwen2.5:1.5b-sixhour; model revision=14bdd46019879e18cd27f21b46e2c377bf801e827e2abb46be2daedcedaf0f64; development_only=True; incremental=False; complete=True; total LLM calls=2; errors=0; tokens=8655. Four controlled development cases; serving batch selection only. Expected labels are excluded from all requests.

Case / carrier	Whole set exact	Tuple precision	Tuple recall	False / omitted	Errors eligible for solver	Escalated
development batch / template	False	—	—	— / —	—	—
development batch / prose	False	—	—	— / —	—	—

Case / carrier / batch	Exact / false / omitted rules	Calls	Seconds	Request result
1 / template / 4	0 / 4 / 4	1	63.954	response received; see exact-rule counts
2 / prose / 4	0 / 4 / 4	1	57.787	response received; see exact-rule counts

These exact-rule counts and durations come from the recorded development cases. Failed requests can have recorded omissions without a validated returned-rule set; missing false-rule/precision fields remain unmeasured. Serving-model selection uses these development samples, which are excluded from inventory-policy comparisons.

Deterministic-template results measure the parser control. Controlled LLM corpus results measure that recorded carrier/case set; neither is a validated natural-language supplier population or evidence that pending M5 studies finished.



Actual development extraction timings. Receiving a response does not establish semantic correctness; exact-rule results are reported above. Different models and batch sizes are separate cases, without pooling, extrapolated completion promises or policy-benefit claims.

Actual live-model grounding and document coverage

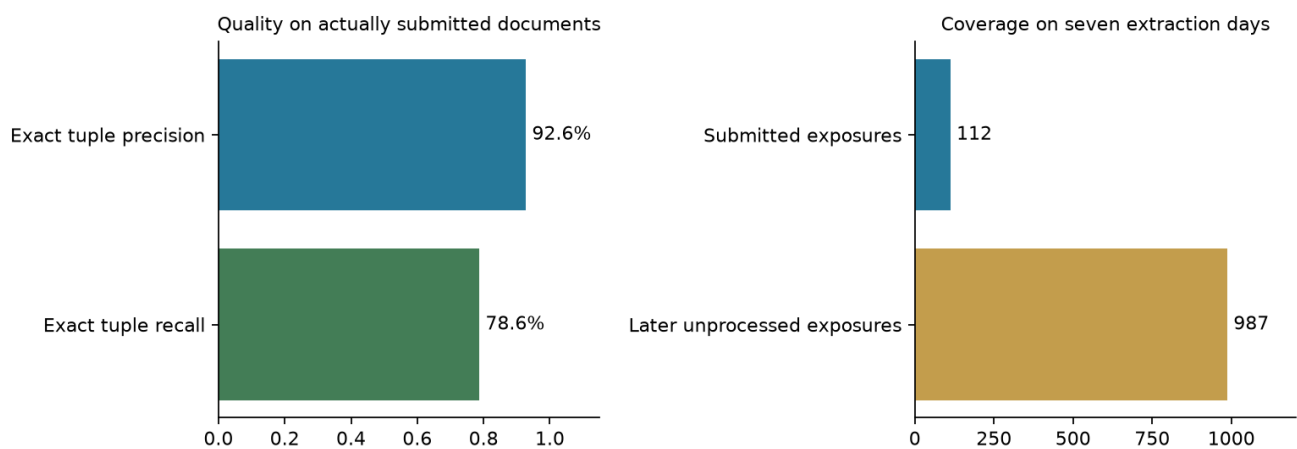
The independent captured-request assessment validates the actual 8-run grid and 16 decision days, with SQLite chains and selected object hashes verified. There were 30 recorded requests: 28 supplier extractions and 2 state triage requests; 112739 tokens (99660 input, 13079 output), 807.8 seconds summed request time, and 0 recorded request errors. API success and schema validity remain distinct from semantic accuracy.

Expected tuples are recovered only from documents actually present in each captured model request. The assessor authenticates source payload SHA-256, strictly inverts the known public prose renderer and requires exact re-rendering before comparing all eleven constraint fields. It does not recall the model or supply it with labels. Confidence, constraint identifiers and provenance are outside the eleven-field tuple score. Exact precision is exact returned tuples divided by returned unique tuples; recall uses expected actually submitted tuples. Later unprocessed documents contribute to coverage, not the submitted-omission denominator.

Conditional grounding measure		Actual value	
Proven submitted / expected tuples		112 / 112	
Exact / false returned tuples		88 / 7	
Omitted submitted tuples		24	
Exact tuple precision / recall		0.92632 / 0.78571	
Unit mismatches / duplicate / unknown-source extras		7 / 0 / 0	
Unproven sources / invalid response requests		0 / 0	
Available / submitted / later unprocessed exposures		1099 / 112 / 987	
Extraction days / days with no extraction		7 / 1	
Decision case (M5 index)	Submitted / available	Calls	Exact / false / omitted
B10 / derived_field_collapse / 1830	16 / 157	4	13 / 1 / 3

Decision case (M5 index)	Submitted / available	Calls	Exact / false / omitted
B10 / derived_field_collapse / 1831	0 / 157	0	not assessed: no extraction
B10 / normal / 1830	16 / 157	4	13 / 1 / 3
B10 / normal / 1831	16 / 157	4	12 / 1 / 4
B9 / derived_field_collapse / 1830	16 / 157	4	13 / 1 / 3
B9 / derived_field_collapse / 1831	16 / 157	4	12 / 1 / 4
B9 / normal / 1830	16 / 157	4	13 / 1 / 3
B9 / normal / 1831	16 / 157	4	12 / 1 / 4

Actual local-model grounding: conditional quality and early-abort coverage



Captured evaluation requests only. Repeated document exposures are not independent trials; conditional extraction accuracy does not imply full-document coverage or successful replenishment.

Data, design and interpretation limits

M5 supplies historical observed sales, calendar and prices. Sales are used as an exogenous demand proxy and may be censored by historical stockouts; latent demand is not recovered. Inventory, replenishment, supplier documents, lead times, capacity, approval, faults and operational constraints are simulated. Simulated run costs and harm outcomes are not measured retailer operations.

The prepared data are selected panels, not all 30,490 bottom-level M5 series or the official competition evaluation. Fault rates are synthetic and uncalibrated unless a configuration explicitly records calibration. Simulated delayed oracle approval is a reviewer model, not a human audit study. Findings are conditional on this panel, window, solver, forecast training, fault definitions and approval model.

Prepared panel	Serie s	Historica l days	Selection	Syntheti c sales?
data/processed/m5	30	1941	{"items": ["FOODS_1_001", "FOODS_2_001", "FOODS_3_001"], "stores": "all"}	False
data/processed/m5_it em1	10	1941	{"items": ["FOODS_1_001"], "stores": "all"}	False
data/processed/m5_s moke	2	1941	{"items": ["FOODS_1_001"], "stores": ["CA_1", "CA_2"]}	False

The archived main stage-1 design requires 480 runs: four deterministic policies × four fault scenarios × thirty independent seeds × one origin. It uses 28 decision days per run. The shortened study has its own registered seed count and horizon. Other stages retain their own panel, carrier and horizon; their cost totals must not be pooled as interchangeable replications. The remaining scenarios and any policies absent from registered configs

are unrun in this report.

Policy	Actual architecture
B1	Deterministic seasonal-naive forecast + order-up-to control
B10	Typed LLM roles + critic + per-decision evidence gate
B2	Trained LightGBM quantile forecast + (s,S) control
B3	Trained native GRU/negative-binomial forecast + stochastic MILP
B4	B3 + deterministic evidence gate; no LLM
B5	Chronos forecast adapter; requires separate weights
B6	Single generalist LLM + shared numerical tools
B7	LLM roles with free-form shared messages + validated action boundary
B8	Typed LLM roles without critic
B9	Typed LLM roles + critic, fixed autonomy
D0	Deterministic seasonal-forecast optimizer software control; no LLM
D1	D0 + deterministic evidence gate software control; no LLM

The native deep model is a trained GRU with a negative-binomial likelihood, not an exact DeepAR or TFT reproduction. LLM roles share deterministic numerical tools and schema validation. LLM calls, errors and explicit fallback decisions are reported separately; a held decision or deterministic fallback does not prove successful semantic reasoning. A seasonal forecast override changes the baseline and must be read in that stage's config.

Revised dissertation coverage under the reduced study

Source: [dissertation_revised_business_school.html](#); SHA-256: 8ae699667fa02f932f4c2584af0a3895021dc925e3119412bb06b238dc87de1c. The short study supplies descriptive evidence relevant to performance, grounding, safety flags and technical auditability. It does not answer the full architecture/memory, severity, equivalent-run, human-audit or universal-reliability hypotheses. No LLM-only numerical-action arm is implemented. Managerial staged adoption remains discussion, not an empirical organizational outcome.

Managerial discussion: a staged rollout would first establish sales-data lineage and deterministic checks, then introduce structured supplier-rule interpretation and auditable traces, and grant bounded autonomy only where matched validation demonstrates adequate reliability. These are proposed adoption conditions. This study does not measure organizational ROI, real reviewer workload, calibrated human trust or field service improvements.

Forecast backtests

These are supplementary previously generated forecast backtests, not additional runs completed inside the six-hour budget. They do not establish that every earlier forecast-scoring hyperparameter matches the reduced study; each current simulation's resolved configuration is the authority. They are shown separately from current-scope outcome counts and are not used here for model selection or confirmatory inference.

Backtests below come from `results/forecast_*.json`. Training ends before each recorded test origin. Preferred holdouts at origins 1760/1788 end by index 1815, before main-study warmup at 1816. Earlier descriptive backtests at 1800/1828 overlap the main-study window and must not be used for model selection. No model or gate settings were retuned on those evaluation scores. WRMSSE, weighted scaled pinball and CRPS are scores for the selected hierarchy; `full_m5_shape=false` means they are not full official M5 benchmark results. Forecast accuracy alone does not establish replenishment quality or an LLM effect.

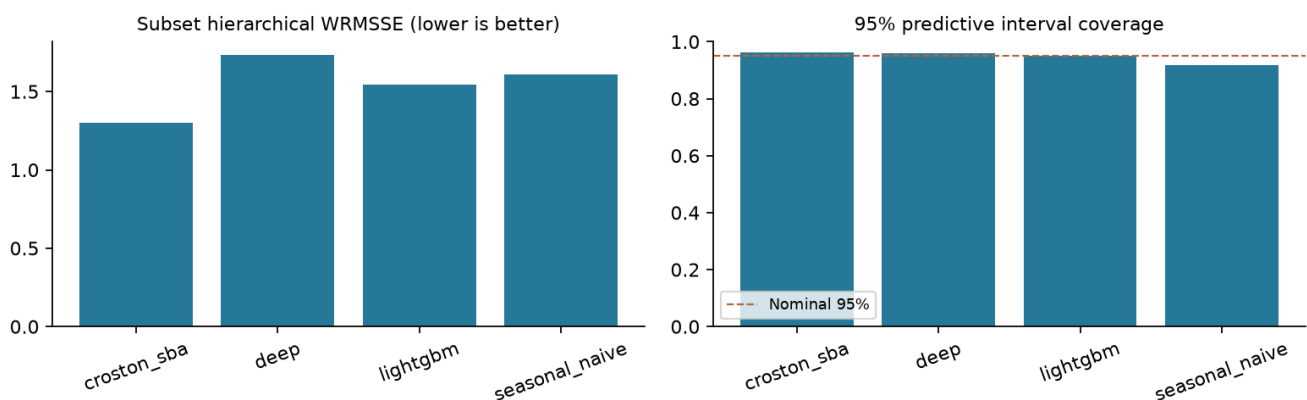
Forecast scores — pre-main-study holdout

Model	Origin / train end	Nodes	WRMSSE	Scaled pinball	CRPS	95% coverage
croston_sba	1760 / 1759	112	1.264	0.585	0.586	0.974
croston_sba	1788 / 1787	112	1.336	0.382	0.630	0.954
deep	1760 / 1759	112	1.445	0.425	0.510	0.977
deep	1788 / 1787	112	2.018	0.476	0.642	0.939
lightgbm	1760 / 1759	112	1.285	0.472	0.540	0.971
lightgbm	1788 / 1787	112	1.803	0.491	0.651	0.931
seasonal_naive	1760 / 1759	112	1.637	0.850	0.784	0.911
seasonal_naive	1788 / 1787	112	1.577	0.461	0.773	0.923

Forecast scores — overlapping descriptive backtest

Model	Origin / train end	Nodes	WRMSSE	Scaled pinball	CRPS	95% coverage
croston_sba	1800 / 1799	112	1.204	0.367	0.628	0.944
croston_sba	1828 / 1827	112	1.428	0.426	0.646	0.946
deep	1800 / 1799	112	1.796	0.407	0.582	0.939
deep	1828 / 1827	112	1.714	0.439	0.621	0.961
lightgbm	1800 / 1799	112	1.493	0.412	0.616	0.939
lightgbm	1828 / 1827	112	1.859	0.534	0.655	0.951
seasonal_naive	1800 / 1799	112	1.538	0.550	0.876	0.886
seasonal_naive	1828 / 1827	112	1.593	0.555	0.808	0.914

Actual forecast backtests; means across recorded origins, no uncertainty intervals



Means over the preferred nonoverlapping holdout origins when available. No confidence intervals or statistical significance claims are inferred from these origins.

Observed simulation outcomes

Tables include completed runs only, separated by stage and policy/scenario. *n* is the number of distinct independent seeds; rolling origins are averaged within seed for reported means. The repository's `harmful_executions` endpoint is a composite flag for true-constraint violations or material action deviation from a same-forecast clean-evidence MILP reference. The code does not require a fault or demonstrate that corrupted evidence caused the deviation. Legitimate classical policies can therefore receive reference-deviation flags in

clean normal runs. Composite flags must not all be interpreted as unsafe or corruption-caused orders; true violations and reference deviations are reported separately. Hard violations evaluate committed plans against the simulator's hidden true planning problem. The simulator can clamp physical transfers to available stock, so a plan violation does not imply negative physical inventory. Totals depend on completed exposure.

Implementation audit's harm-definition caveat: Harmful execution in code is true-constraint violation OR material distance from same-forecast clean-evidence MILP reference. It does not require an injected fault or causal attribution to corrupted evidence. Therefore legitimate clean-state classical-policy/reference differences can count as reference deviations. Fresh baseline_pilot normal rows confirm B1 has 7 deviations/0 violations and B2 has 5 deviations/0 violations. Report the two endpoints separately; do not describe all combined flags as unsafe or corruption-caused orders.

Partial cells can be selected by execution order and are unsuitable for policy ranking or confirmatory inference. This report makes no claim of statistical power, equivalence, significance, or treatment benefit from partial/pilot data. The repository's thirty-seed requirement is a protocol minimum, not proof of adequate power. Zero observed harms is not a universal safety guarantee.

six_hour_agentic_prose — COMPLETE

Output: results/six_hour_agentic_prose; config: results/run_configs/six_hour_agentic_prose.yaml; dataset: data/processed/m5; series attested by completed-run manifest: 30. Seeds configured: 1; origins: 1; start index: 1830; warmup: 14; decision days: 2; approval: oracle with delay 1; forecast override: none recorded. M5 index 0 corresponds to d_1.

Historical origin 0: training data end 2016-01-18 at index 1815 (d_1816); decisions 2016-02-02 to 2016-02-03 at indices 1830-1831 (d_1831-d_1832). Source: data/processed/m5/calendar.csv.

B3/B4 in this prose stage are intentional deterministic-parser comparators. The parser requires the structured RULE carrier and recognizes no rules in controlled prose, so missing required constraint coverage causes holds. They are not structured-rule oracle benchmarks. Separate registered two-day template controls are shown under their own stage; the seven-day numerical template costs are not pooled into this comparison.

Policy / scenario	n seeds / runs	Mean cost	Mean fill	Harm total	True violation / ref deviation	Holds total
B10 / derived_field_collapse	1 / 1	8.513	1.000	0	0 / 0	2
B10 / normal	1 / 1	8.513	1.000	0	0 / 0	2
B3 / derived_field_collapse	1 / 1	8.513	1.000	0	0 / 0	2
B3 / normal	1 / 1	8.513	1.000	0	0 / 0	2
B4 / derived_field_collapse	1 / 1	8.513	1.000	0	0 / 0	2
B4 / normal	1 / 1	8.513	1.000	0	0 / 0	2
B9 / derived_field_collapse	1 / 1	8.513	1.000	0	0 / 0	2
B9 / normal	1 / 1	8.513	1.000	0	0 / 0	2

Policy / scenario	LLM calls	LLM errors	Fallback decisions
B10 / derived_field_collapse	5	0	0
B10 / normal	8	0	0
B3 / derived_field_collapse	0	0	0
B3 / normal	0	0	0
B4 / derived_field_collapse	0	0	0
B4 / normal	0	0	0
B9 / derived_field_collapse	9	0	0

Policy / scenario	LLM calls		LLM errors	Fallback decisions	
B9 / normal	8		0	0	

Policy / scenario	Cycle service	Stockout rate	Bullwhip	Mean inventory	Inventory turns
B10 / derived_field_collapse	1.000	0.000	0.000	237.500	0.232
B10 / normal	1.000	0.000	0.000	237.500	0.232
B3 / derived_field_collapse	1.000	0.000	0.000	237.500	0.232
B3 / normal	1.000	0.000	0.000	237.500	0.232
B4 / derived_field_collapse	1.000	0.000	0.000	237.500	0.232
B4 / normal	1.000	0.000	0.000	237.500	0.232
B9 / derived_field_collapse	1.000	0.000	0.000	237.500	0.232
B9 / normal	1.000	0.000	0.000	237.500	0.232

Completed-run recorded tokens=112,739; LLM calls=30; LLM errors=0; fallback decisions=0; invalid audit chains=0. Counts exclude unfinished runs. Local-model tokens are not assigned a fabricated dollar value.

six_hour_agentic_prose actual proposal-solver diagnostics

Policy / scenario	MILP attempts	Feasible / optimal	Time-limit incumbent	Mean / max gap	Mean seconds
B10 / derived_field_collapse	0	0 / 0	0	— / —	—
B10 / normal	0	0 / 0	0	— / —	—
B3 / derived_field_collapse	0	0 / 0	0	— / —	—
B3 / normal	0	0 / 0	0	— / —	—
B4 / derived_field_collapse	0	0 / 0	0	— / —	—
B4 / normal	0	0 / 0	0	— / —	—
B9 / derived_field_collapse	0	0 / 0	0	— / —	—
B9 / normal	0	0 / 0	0	— / —	—

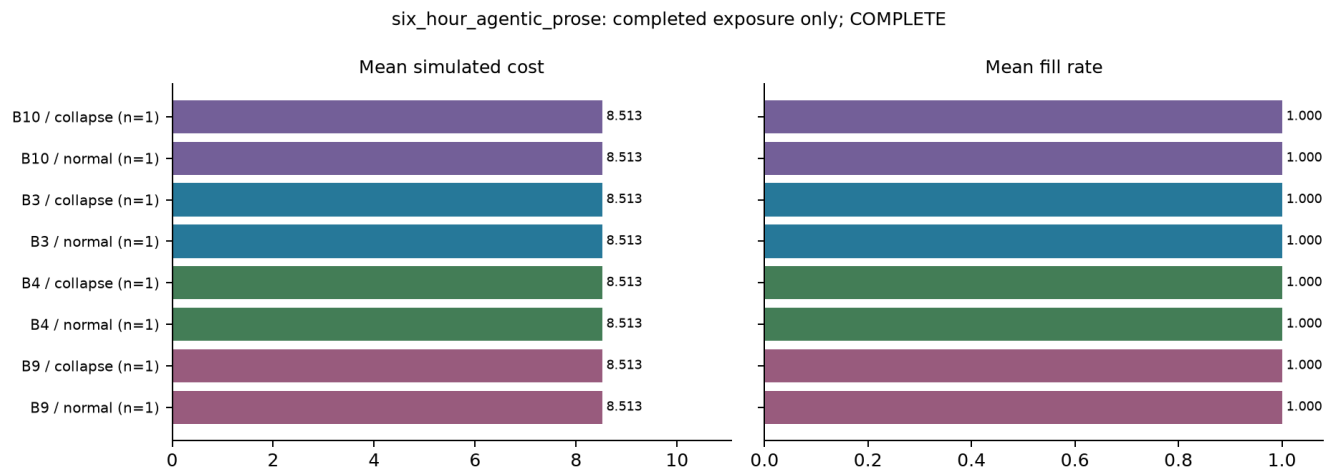
These counts read every available completed-run proposal trace, including verified ZIP objects. A feasible incumbent is not necessarily optimal. Gaps are reported relative fractions (0.001 means 0.1%) over proposals that report a finite gap; missing gaps are not zero. The five-second solver limit can select different incumbents and affect cost comparisons. Held decisions can have solver status not_run; solver feasibility does not establish that a proposal was executed or safe under hidden simulator truth.

Recorded held case	Solver status	First dimensional mismatch / reason excerpt
B10 / derived_field_collapse / day 1830	not_run	FOODS_1_001@CA_3:unit_cost:1830: dimensional mismatch USD
B10 / derived_field_collapse / day 1831	not_run	State hard failure; hold is not a zero-order recommendation
B10 / normal / day 1830	not_run	FOODS_1_001@CA_3:unit_cost:1830: dimensional mismatch USD
B10 / normal / day 1831	not_run	FOODS_1_001@CA_3:unit_cost:1831: dimensional mismatch USD

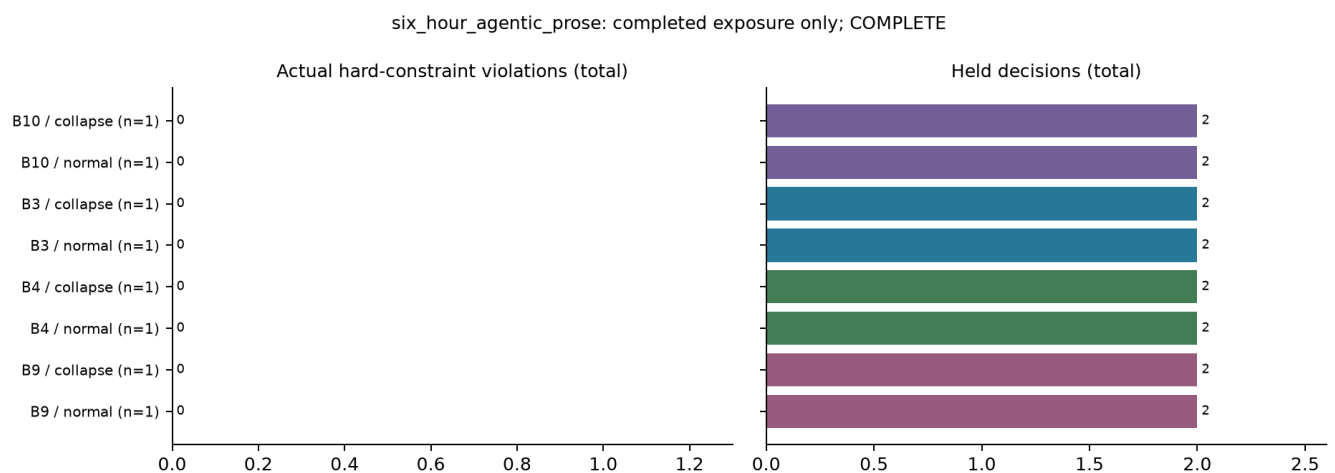
Recorded held case	Solver status	First dimensional mismatch / reason excerpt
B3 / derived_field_collapse / day 1830	not_run	Constraint escalation: FOODS_1_001@CA_1: missing required eligibility; FOODS_1_001@CA_1: missing required lead_time; FOODS_1_001@CA_1: missing required moq; FOODS_1_001@C
B3 / derived_field_collapse / day 1831	not_run	Constraint escalation: FOODS_1_001@CA_1: missing required eligibility; FOODS_1_001@CA_1: missing required lead_time; FOODS_1_001@CA_1: missing required moq; FOODS_1_001@C
B3 / normal / day 1830	not_run	Constraint escalation: FOODS_1_001@CA_1: missing required eligibility; FOODS_1_001@CA_1: missing required lead_time; FOODS_1_001@CA_1: missing required moq; FOODS_1_001@C
B3 / normal / day 1831	not_run	Constraint escalation: FOODS_1_001@CA_1: missing required eligibility; FOODS_1_001@CA_1: missing required lead_time; FOODS_1_001@CA_1: missing required moq; FOODS_1_001@C
B4 / derived_field_collapse / day 1830	not_run	Constraint escalation: FOODS_1_001@CA_1: missing required eligibility; FOODS_1_001@CA_1: missing required lead_time; FOODS_1_001@CA_1: missing required moq; FOODS_1_001@C
B4 / derived_field_collapse / day 1831	not_run	State hard failure; hold is not a zero-order recommendation
B4 / normal / day 1830	not_run	Constraint escalation: FOODS_1_001@CA_1: missing required eligibility; FOODS_1_001@CA_1: missing required lead_time; FOODS_1_001@CA_1: missing required moq; FOODS_1_001@C
B4 / normal / day 1831	not_run	Constraint escalation: FOODS_1_001@CA_1: missing required eligibility; FOODS_1_001@CA_1: missing required lead_time; FOODS_1_001@CA_1: missing required moq; FOODS_1_001@C
B9 / derived_field_collapse / day 1830	not_run	FOODS_1_001@CA_3:unit_cost:1830: dimensional mismatch USD
B9 / derived_field_collapse / day 1831	not_run	FOODS_1_001@CA_3:unit_cost:1831: dimensional mismatch USD
B9 / normal / day 1830	not_run	FOODS_1_001@CA_3:unit_cost:1830: dimensional mismatch USD
B9 / normal / day 1831	not_run	FOODS_1_001@CA_3:unit_cost:1831: dimensional mismatch USD

Dimensional mismatch is a specific invalid supplied rule. Missing-required-field messages can also reflect documents never processed after early abort; they are not all model extraction omissions. Submitted-document grounding accuracy is evaluated only against captured input batches, with unprocessed coverage shown separately. Item aliases are permitted by verification; a dropped store suffix is a wrong tuple/scope, not automatically an unknown-entity runtime error.

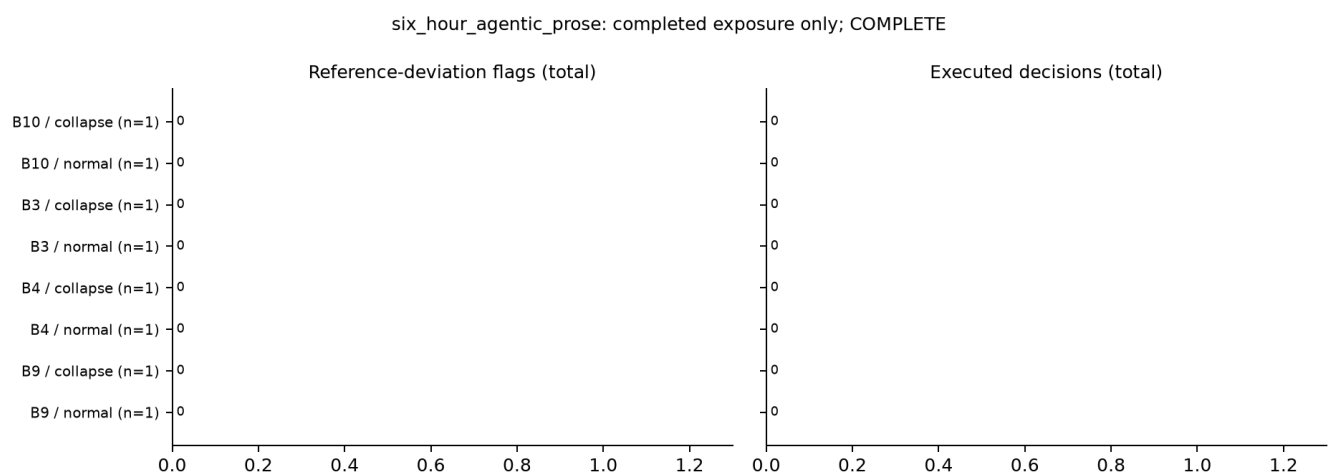
The repository also wrote study_summary.json with replication-level cost/CVaR and utility. It is preserved in report_data.json. It may lag live worker results; this report's tables are refreshed from complete per-run artifacts. Run-level daily CVaR is not substituted for replication-level tail risk.



Descriptive simulated tradeoffs at the completed exposure (n is simulation seeds). Actual hard violations, holds, reference deviations and executed decisions remain distinct endpoints. Short-window inventory accounting and incomplete cells limit policy comparisons; no LLM savings or significance is inferred.



Descriptive simulated tradeoffs at the completed exposure (n is simulation seeds). Actual hard violations, holds, reference deviations and executed decisions remain distinct endpoints. Short-window inventory accounting and incomplete cells limit policy comparisons; no LLM savings or significance is inferred.



Descriptive simulated tradeoffs at the completed exposure (n is simulation seeds). Actual hard violations, holds, reference deviations and executed decisions remain distinct endpoints. Short-window inventory accounting and incomplete cells limit policy comparisons; no LLM savings or significance is inferred.

six_hour_numeric_30seeds — COMPLETE

Output: results/six_hour_numeric_30seeds; config: results/six_hour_numeric_30seeds/resolved_config.json; dataset: data/processed/m5; series attested by completed-run manifest: 30. Seeds configured: 30; origins: 1; start index: 1830; warmup: 14; decision days: 7; approval: oracle with delay 1; forecast override: none recorded. M5 index 0 corresponds to d_1.

Historical origin 0: training data end 2016-01-18 at index1815 (d_1816); decisions 2016-02-02 to 2016-02-08 at indices1830-1836 (d_1831-d_1837). Source: data/processed/m5/calendar.csv.

Policy / scenario	n seeds / runs	Mean cost	Mean fill	Harm total	True violation / ref deviation	Holds total
B1 / derived_field_collapse	30 / 30	621.067	0.928	179	0 / 179	0
B1 / feed_gap	30 / 30	443.184	0.908	210	0 / 210	0
B1 / normal	30 / 30	399.847	0.920	209	0 / 209	0
B2 / derived_field_collapse	30 / 30	548.971	0.852	122	0 / 122	0
B2 / feed_gap	30 / 30	440.066	0.772	131	0 / 131	0
B2 / normal	30 / 30	399.629	0.810	161	0 / 161	0
B3 / derived_field_collapse	30 / 30	497.046	0.937	78	0 / 78	0
B3 / feed_gap	30 / 30	425.570	0.903	100	63 / 37	0
B3 / normal	30 / 30	375.362	0.940	10	0 / 10	0
B4 / derived_field_collapse	30 / 30	369.423	0.905	6	0 / 6	90
B4 / feed_gap	30 / 30	367.796	0.905	5	0 / 5	90
B4 / normal	30 / 30	374.000	0.939	11	0 / 11	0

Policy / scenario	LLM calls	LLM errors	Fallback decisions
B1 / derived_field_collapse	0	0	0
B1 / feed_gap	0	0	0
B1 / normal	0	0	0
B2 / derived_field_collapse	0	0	0
B2 / feed_gap	0	0	0
B2 / normal	0	0	0
B3 / derived_field_collapse	0	0	0
B3 / feed_gap	0	0	0
B3 / normal	0	0	0
B4 / derived_field_collapse	0	0	0
B4 / feed_gap	0	0	0
B4 / normal	0	0	0

Policy / scenario	Cycle service	Stockout rate	Bullwhip	Mean inventory	Inventory turns
B1 / derived_field_collapse	0.490	0.023	52.979	255.505	0.804
B1 / feed_gap	0.319	0.033	2.407	207.210	0.973
B1 / normal	0.400	0.028	1.957	206.676	0.989
B2 / derived_field_collapse	0.281	0.046	47.444	226.205	0.835
B2 / feed_gap	0.181	0.074	6.143	180.548	0.954
B2 / normal	0.195	0.064	2.026	183.952	0.981

Policy / scenario	Cycle service	Stockout rate	Bullwhip	Mean inventory	Inventory turns
B3 / derived_field_collapse	0.514	0.021	20.964	220.724	0.943
B3 / feed_gap	0.367	0.040	2.987	177.767	1.124
B3 / normal	0.510	0.023	3.547	174.700	1.192
B4 / derived_field_collapse	0.381	0.037	11.464	172.667	1.163
B4 / feed_gap	0.390	0.036	11.454	172.743	1.163
B4 / normal	0.495	0.023	3.550	174.343	1.194

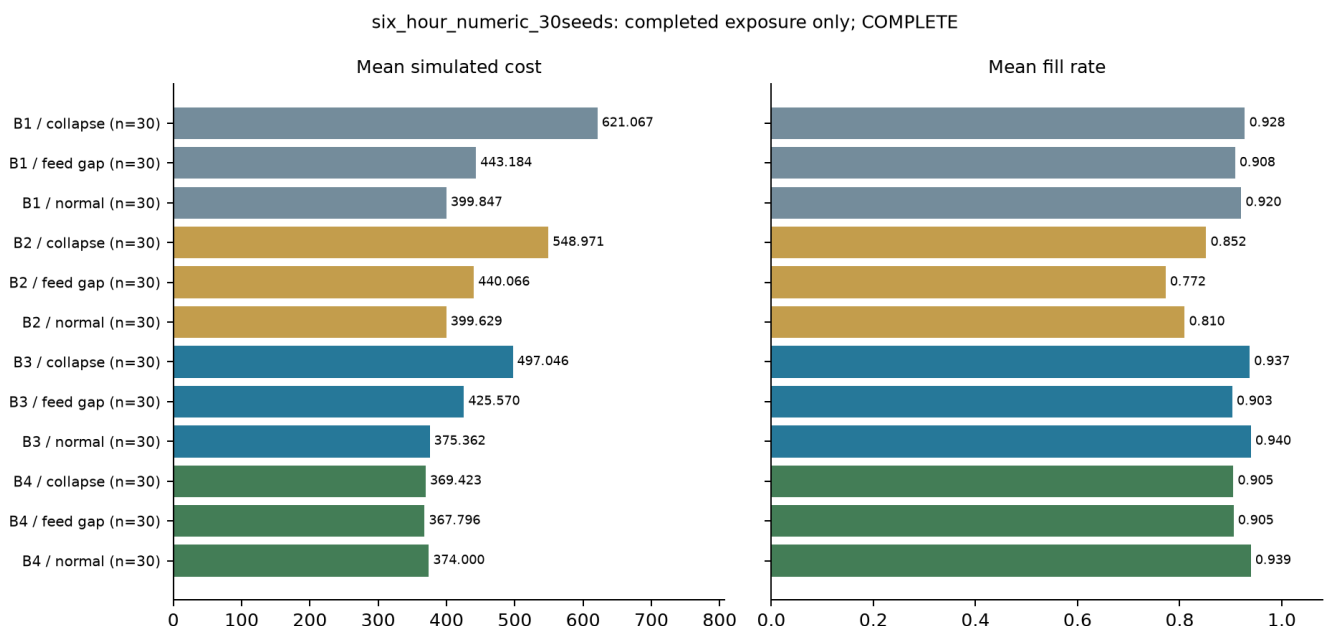
Completed-run recorded tokens=0; LLM calls=0; LLM errors=0; fallback decisions=0; invalid audit chains=0. Counts exclude unfinished runs. Local-model tokens are not assigned a fabricated dollar value.

six_hour_numeric_30seeds actual proposal-solver diagnostics

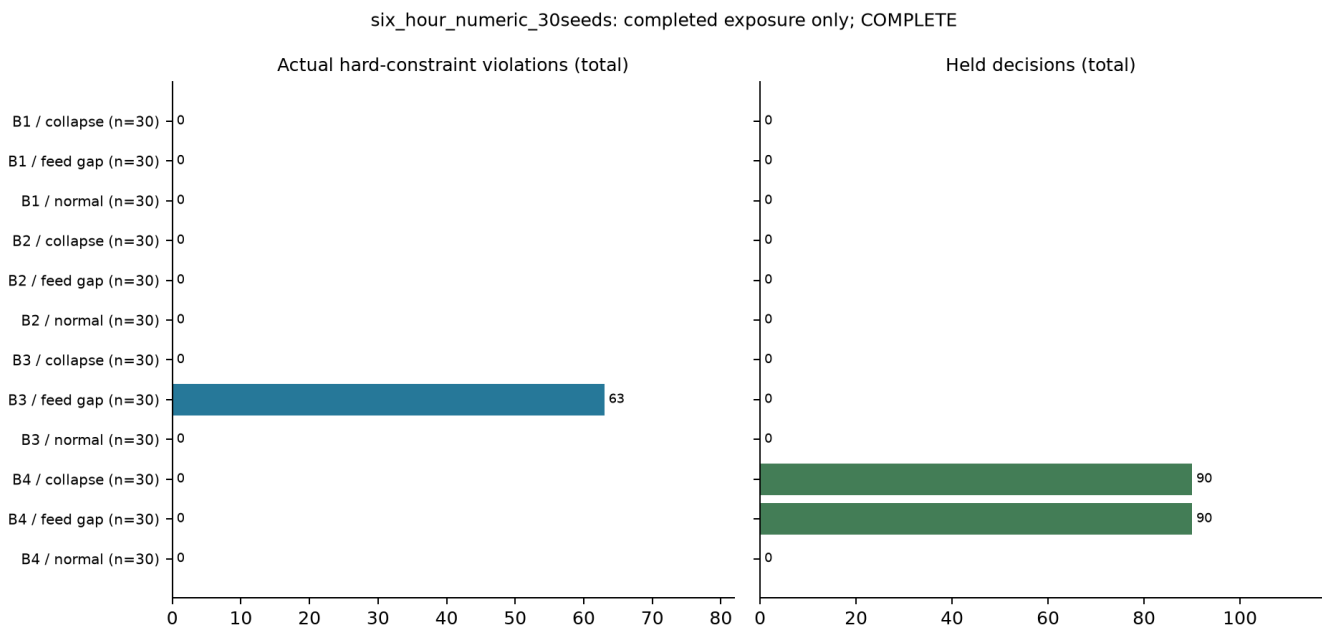
Policy / scenario	MILP attempts	Feasible / optimal	Time-limit incumbent	Mean / max gap	Mean seconds
B3 / derived_field_collapse	210	210 / 210	0	0.00021 / 0.00100	0.369
B3 / feed_gap	210	210 / 210	0	0.00043 / 0.00100	0.917
B3 / normal	210	210 / 210	0	0.00044 / 0.00100	1.035
B4 / derived_field_collapse	120	120 / 120	0	0.00038 / 0.00100	0.862
B4 / feed_gap	120	120 / 120	0	0.00040 / 0.00100	0.840
B4 / normal	210	210 / 210	0	0.00044 / 0.00100	0.990

These counts read every available completed-run proposal trace, including verified ZIP objects. A feasible incumbent is not necessarily optimal. Gaps are reported relative fractions (0.001 means 0.1%) over proposals that report a finite gap; missing gaps are not zero. The five-second solver limit can select different incumbents and affect cost comparisons. Held decisions can have solver status not_run; solver feasibility does not establish that a proposal was executed or safe under hidden simulator truth.

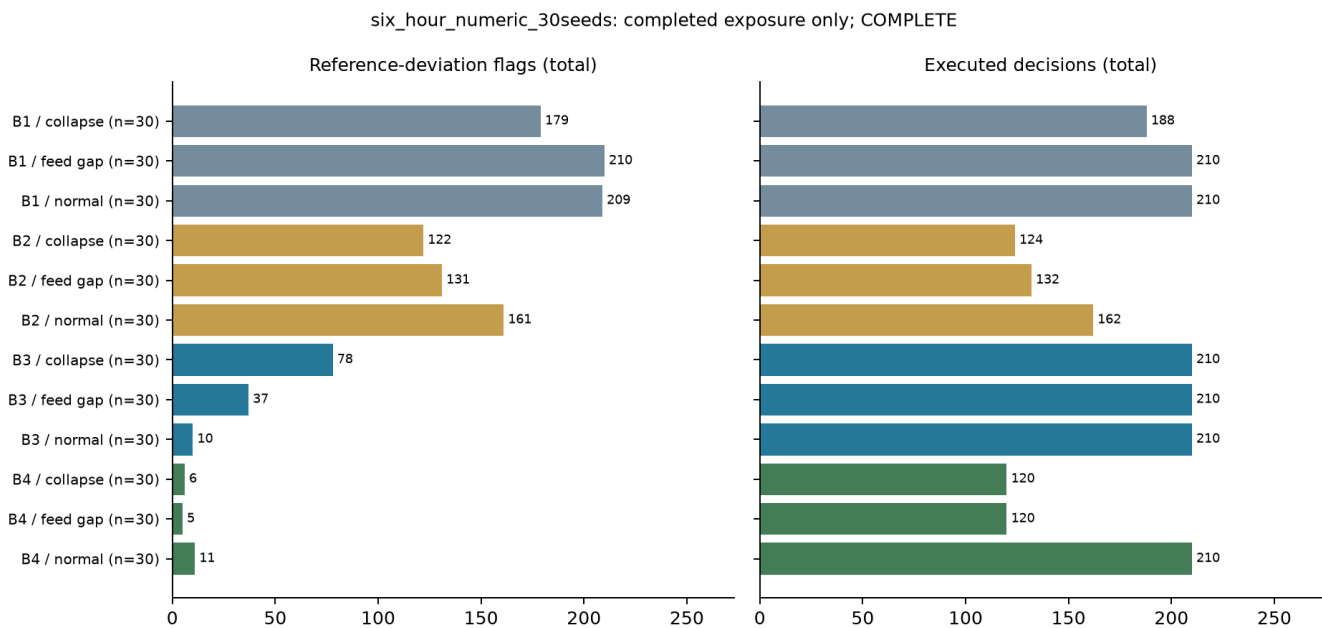
The repository also wrote study_summary.json with replication-level cost/CVaR and utility. It is preserved in report_data.json. It may lag live worker results; this report's tables are refreshed from complete per-run artifacts. Run-level daily CVaR is not substituted for replication-level tail risk.



Descriptive simulated tradeoffs at the completed exposure (n is simulation seeds). Actual hard violations, holds, reference deviations and executed decisions remain distinct endpoints. Short-window inventory accounting and incomplete cells limit policy comparisons; no LLM savings or significance is inferred.



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six_hour_structured_controls — COMPLETE

Output: results/six_hour_structured_controls; config: results/run_configs/six_hour_structured_controls.yaml; dataset: data/processed/m5; series attested by completed-run manifest: 30. Seeds configured: 1; origins: 1; start index: 1830; warmup: 14; decision days: 2; approval: oracle with delay 1; forecast override: none recorded. M5 index 0 corresponds to d_1.

Historical origin 0: training data end 2016-01-18 at index1815 (d_1816); decisions 2016-02-02 to 2016-02-03 at indices1830-1831 (d_1831-d_1832). Source: data/processed/m5/calendar.csv.

Policy / scenario	n seeds / runs	Mean cost	Mean fill	Harm total	True violation / ref deviation	Holds total
B3 / derived_field_collapse	1 / 1	178.272	1.000	0	0 / 0	0
B3 / normal	1 / 1	97.066	1.000	0	0 / 0	0
B4 / derived_field_collapse	1 / 1	67.673	1.000	0	0 / 0	1
B4 / normal	1 / 1	96.499	1.000	0	0 / 0	0

Policy / scenario	LLM calls	LLM errors	Fallback decisions
B3 / derived_field_collapse	0	0	0
B3 / normal	0	0	0
B4 / derived_field_collapse	0	0	0
B4 / normal	0	0	0

Policy / scenario	Cycle service	Stockout rate	Bullwhip	Mean inventory	Inventory turns
B3 / derived_field_collapse	1.000	0.000	10.467	231.000	0.238
B3 / normal	1.000	0.000	1.526	224.000	0.246
B4 / derived_field_collapse	1.000	0.000	5.817	231.000	0.238
B4 / normal	1.000	0.000	1.526	225.000	0.244

Completed-run recorded tokens=0; LLM calls=0; LLM errors=0; fallback decisions=0; invalid audit chains=0. Counts exclude unfinished runs. Local-model tokens are not assigned a fabricated dollar value.

six_hour_structured_controls actual proposal-solver diagnostics

Policy / scenario	MILP attempts	Feasible / optimal	Time-limit incumbent	Mean / max gap	Mean seconds
B3 / derived_field_collapse	2	2 / 2	0	0.00077 / 0.00083	0.911
B3 / normal	2	2 / 2	0	0.00057 / 0.00083	1.567
B4 / derived_field_collapse	1	1 / 1	0	0.00083 / 0.00083	1.596
B4 / normal	2	2 / 2	0	0.00080 / 0.00083	1.221

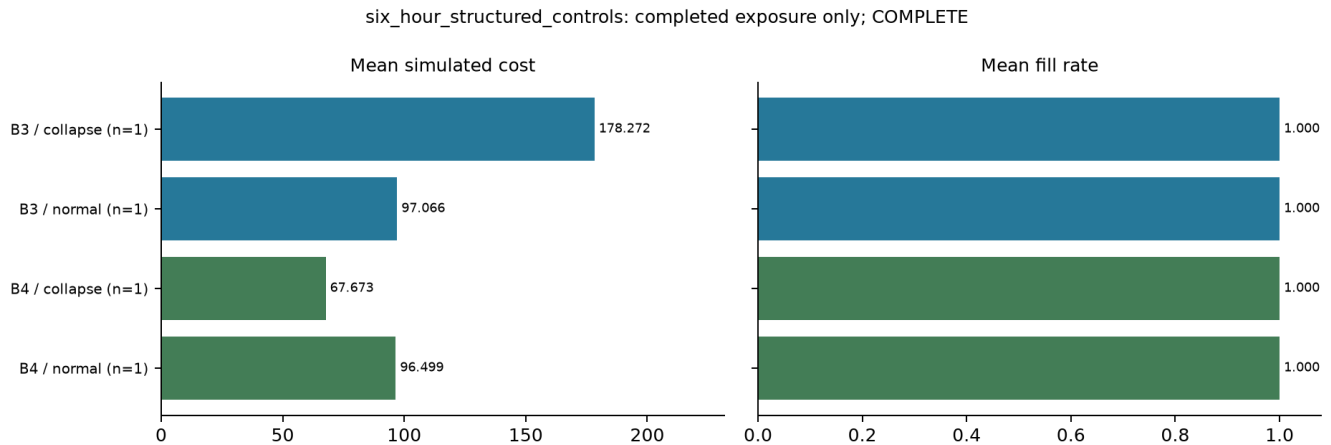
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Recorded held case	Solver status	First dimensional mismatch / reason excerpt
B4 / derived_field_collapse / day 1831	not_run	State hard failure; hold is not a zero-order recommendation

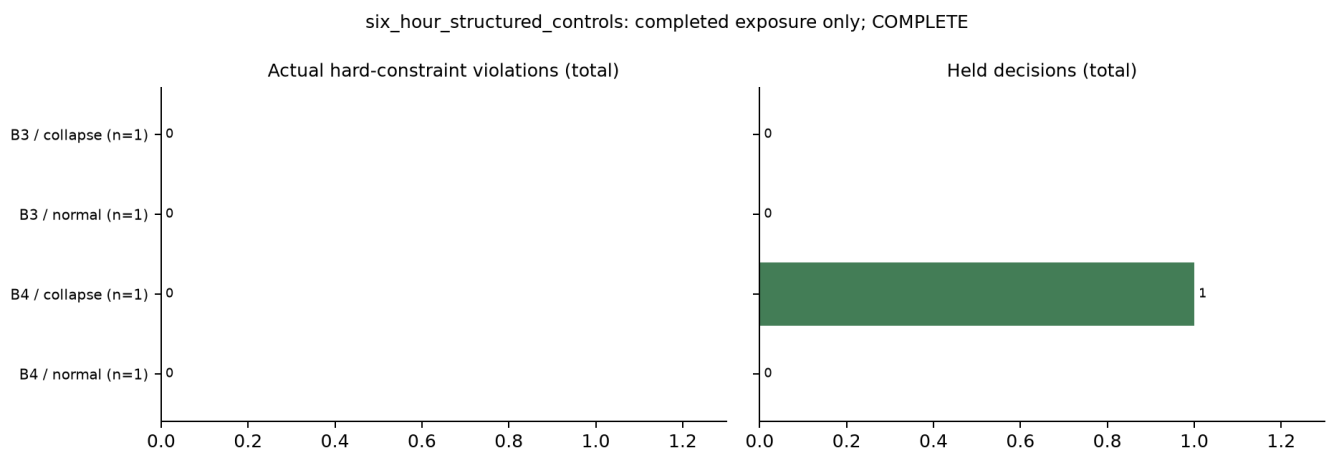
Dimensional mismatch is a specific invalid supplied rule. Missing-required-field messages can also reflect documents never processed after early abort; they are not all model extraction omissions. Submitted-document grounding accuracy is evaluated only against captured input batches, with unprocessed coverage shown

separately. Item aliases are permitted by verification; a dropped store suffix is a wrong tuple/scope, not automatically an unknown-entity runtime error.

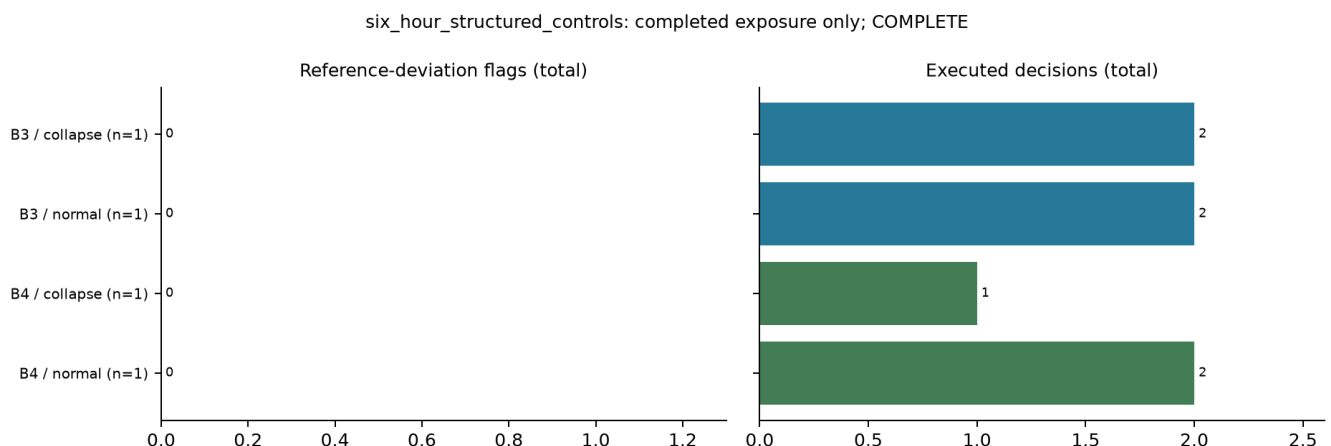
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Descriptive analysis only

No significance tests or confirmatory hypothesis decisions are generated for the six-hour scope. Run and seed counts are shown as actually completed. Grounding sample proportions are descriptive, with no independence or power claim; equal seeds match random streams but do not turn a small pilot into a confirmatory study. Earlier long-study paired comparisons are preserved in their original report and are excluded here.

The numerical extension was registered after the initial completed cohort was available and was justified by runtime capacity, not as a confirmatory dissertation design. The fixed B4-minus-B3 cost/fill/actual-hard-violation intervals below are explicitly exploratory: thirty matched simulation seeds, one origin, seven days, 5,000 paired percentile bootstrap resamples with seed42 and unadjusted 95% intervals. No p-values, significance-driven superiority or statistical-power claims are made. These intervals condition on this panel and fixed trained forecast, and do not establish zero-event safety, equivalence or external validity.

Scenario	Metric	Pairs	B4 minus B3 mean	Exploratory 95% interval
normal	cost	30	-1.36171	[-4.11611, 1.24449]
normal	fill_rate	30	-0.00015	[-0.00182, 0.00152]
normal	hard_violations	30	0.00000	[0.00000, 0.00000]
derived_field_collapse	cost	30	-127.62342	[-141.93711, -112.62178]
derived_field_collapse	fill_rate	30	-0.03212	[-0.04439, -0.02182]
derived_field_collapse	hard_violations	30	0.00000	[0.00000, 0.00000]
feed_gap	cost	30	-57.77395	[-76.57303, -38.33628]
feed_gap	fill_rate	30	0.00258	[-0.01288, 0.01697]
feed_gap	hard_violations	30	-2.10000	[-2.40000, -1.76667]

Validation and reproducibility checks

Independent final numerical source audit: valid=True, verified at 2026-10-07T21:28:34.281654+00:00. It checked 360 original source runs, 2520 traces, 27180 SQLite chain events and 33442 original objects. Exact disjoint coverage, immutable source hashes, all-worker trained-model equality and every archive entry's bytes/hash/size were verified. No original run manifests were cloned. This establishes source and artifact integrity, not causal grounding faithfulness or treatment superiority.

Lossless verified archives occupy 108384628 bytes for 994587177 bytes of original objects. 28612 currently removed loose objects remain recoverable; packing states at audit time: {"packed": 358, "restored": 2}. Full replay tools need a restore unless they implement the verified ZIP reader. The standalone audit and its SHA-256 are included in the report input inventory.

Recorded existing test-suite validation: passed=97, failed=0, skipped=1. Skip reason: optional anthropic package not installed. This is the recorded validation manifest; report generation does not rerun the suite.

Dataset acquisition provenance: scripts/fetch_m5.py downloaded the Nixtla public mirror and restored missing id/day labels without changing sales/prices/date values

Local Ollama recorded setup: {"binary_sha256": "c94aa4156b3d13e64ebc2efe5ea53f015384c882be776e6695cfb37fb180d5ad", "model_digest": "bd9b617abc6c2ec05ebeac915201900320377281168b3949032799cc8f20d51d", "model_identity": "results/run_configs/ollama_model_identity.json", "model_loaded_for_inference": false, "registry_connectivity": "Official manifest, model blob and other blobs downloaded over TLS; SHA256 and sizes verified; native pull succeeded", "version": "0.40.0"}

```
results/logs/ml_m5_replay.json: {"action_matches": true, "chain_valid": true, "decision_id":
"B4-normal-seed7-origin0:day1800", "held": false, "independent_violations": [], "mode": "cached external
evidence/tool replay; does not re-call the LLM or retrain forecasts", "objective_difference": 0.0,
"problem_reconstruction_matches": true, "problem_wire_hash_matches": true, "recorded_action_hash":
"cba7105ab172cb685fd367cb5b756f4a69a436786d8379d76bea884ec2644099", "replay_status": "exact action
match", "replayed_action_hash":
"cba7105ab172cb685fd367cb5b756f4a69a436786d8379d76bea884ec2644099", "state_certificate_matches":
true}
```

```
results/logs/ml_m5_trace_audit.json: {"artifacts_hash_verified": true, "completeness": 1.0,
"consumption_precision": 1.0, "consumption_recall": 0.9, "note": "Consumption is instrumentation evidence, not
proof that every consumed field was decisive. No neural chain-of-thought claims."}
```

Selected six-hour replay/faithfulness evidence: Actual completed six-hour numerical and live local-LLM traces; four deliberately selected decision cases. Four deliberately selected cases, not a random or full solver-replay audit. Cached action replay uses the observed problem; separate deterministic feasibility recheck uses post-decision evaluator true_problem with comparison lineage aligned as in production evaluator. Simulator physical fulfillment clamps infeasible transfer quantities, so violations describe committed plans rather than negative physical stock.

Selected policy / day	Replay result	Chain / state match	Completeness	Consumption precision / recall
B3 / 1833	exact action match	True / True	1.000	1.000 / 0.900
B4 / 1833	reproduced fail-closed held decision	True / True	1.000	0.667 / 0.800
B10 / 1830	reproduced fail-closed held decision	True / True	1.000	1.000 / 0.889
B10 / 1831	reproduced fail-closed held decision	True / True	1.000	0.667 / 0.800

The B3 feed-gap day1833 case reproduces its recorded observed problem/action/hash while the simulator daily file records a hard violation of the committed plan. An empty independent_violations list against the observed planning problem is not proof of safety under hidden simulator truth. Ground truth is stored after the decision inside reader.load(trace['evaluation'])['true_problem'], outside trace.references; it is distinct from the observed problem and was not supplied to the model. Independent checks of that post-decision truth are preserved in this selected audit artifact. Physical transfers can be clamped to available stock, so a committed-plan violation is not a negative-inventory claim. The B4 hold and two selected live B10 collapse holds reproduce fail-closed behavior. These four deliberately selected cases are not a random sample or full exact-action replay audit. Consumption scores below one remain reported; structural deletion of a required forecast blocks reconstruction but does not establish semantic context faithfulness or causal importance.

Sample replay and trace-audit results validate their selected recorded decisions; they are not audits of every pending study run or re-executions of an LLM. Live functional and grounding validation are reported separately from these cached replay checks.

Reproducibility and artifact provenance

Current checkout commit: bbe3bbcad24291994fbc70fb329f165d2d97bb96. Tracked working-tree changes at report time: none. This attests the current checkout only; original per-run manifests do not necessarily contain a code SHA. Run configuration, environment and data evidence are taken from their actual artifacts. M5_six_hour_study_report_input_hashes.json records SHA-256 for every input file used by this report.

Retained earlier launch protocol recorded code commit: bbe3bbcad24291994fbc70fb329f165d2d97bb96; model: ega-llama3.2:3b-cloud30; model revision: bd9b617abc6c2ec05ebeac915201900320377281168b3949032799cc8f20d51d. Models selected for the bounded study are identified separately in their current identity and frozen run configs. Findings from earlier local or historical hosted models cannot be inherited by a changed-model run.

Launch protocol's recorded scope: No model selection or gate retuning on evaluation windows Current required/auxiliary stage settings in the execution table take precedence if the registered scope has since expanded.

Launch protocol's recorded scope: Local 3b model is different from repository Cerebras study; cannot inherit historical provider findings Current required/auxiliary stage settings in the execution table take precedence if the registered scope has since expanded.

Launch protocol's recorded scope: M5 historical sales proxy; inventory, supply, documents and perturbations simulated Current required/auxiliary stage settings in the execution table take precedence if the registered scope has since expanded.

Launch protocol's recorded scope: Full 6120-run deterministic config and Chronos foundation baseline are separate extensions; not silently treated as executed Current required/auxiliary stage settings in the execution table take precedence if the registered scope has since expanded.

Raw M5 source	SHA-256 recorded by preparation manifest
calendar.csv	f9295292394975908e04ac77898c2304cbd499a2c466b2b9ce6f04079675bf16
sales_train_evaluation.csv	4b4a47c44c38380d2a9168216fea8c9ff2f31b1ddb772f8a0995952a038b8aa0
sell_prices.csv	5c7cf045b1abdbdc8d5007809c40b3682c5216075937f93e2cef01caddc4a30d9

The raw-source hashes above are recorded by the prepared-data manifests; report generation does not independently rehash large raw files. Per-panel catalog hashes and complete safe config snapshots are included in the HTML/Markdown and report_data.json. The input hash inventory permits detecting changes between report snapshots.

Lossless object packing and full-trace access

Current configured run artifacts with recognized packing manifests: 393. Packing stores the full original content-addressed JSON object inventory in artifacts/objects.zip and keeps the first three trace/certificate/proposal objects loose for standard reports. Missing loose copies of archived objects are not missing data when the inventory, archive and provenance verify. Summaries, daily rows, trace indices and the SQLite audit chain remain separate artifacts.

Stage	Packed/restored runs	Trace decisions available	Loose trace decisions	ZIP logical bytes
main_study_stage1	20	560	60	48,330,604
llm_prose_reference_30series	1	28	3	3,469,164
six_hour_agentic_prose	8	16	16	668,635
six_hour_numeric_30seeds	360	2520	1088	108,384,628
six_hour_structured_controls	4	8	8	357,870

The report checks current archive SHA/size, exact ZIP-member inventory, original inventory hash and unchanged run provenance. Archive SHA checks are reused only for the same file identity, size, modification time and expected checksum. Packing records attest byte-by-byte verification of original objects and the actual audit chain before removal. Full trace indices remain counted even when only three trace objects are loose.

Packing helper validation used a copied fixture, not experiment-output modification: original objects=367; retained after packing=9; archive bytes=2417610; actual outputs modified=False. These helper checks do not imply that all real study runs have been packed or independently audited.

This report's selected-object ZIP reader was checked on a copied fixture: archive/loose JSON equality=True; unknown reference refused=True; whole-run restoration performed by that check=False; actual experiment outputs modified=False.

Selected JSON can be read directly with this generator's PackedArtifactReader, which validates the requested object's original SHA-256 and byte length without whole-run restoration. Existing full replay, audit, grounding-artifact inspection and human-packet tools need restoration unless they explicitly use such a ZIP reader. Restore the CURRENT run directory with `/workspace/MasterDissertation/.venv/bin/python /workspace/tools/m5_pack_artifacts.py --restore <CURRENT_RUN_DIRECTORY>`; use the current merged location rather than a former worker path. This report performs no restoration or packing.

Stage	Input config SHA-256
m5_smoke	107d0cc30a834c4a04b45ef40b9ce9824924a27a8564bb740d6f2edf80b263e2
ml_m5_smoke	d64f153ca37d80bd161c42a02fd9693ee0568b674f7e6e409dd98c353a998bac
baseline_pilot	83955a0d49aa3c1ea79915d70ed22dee7fdbfcedf383cacaee112815a95fbff7d
main_study_stage1	b7df6c95d2d9dbfd9498ff314410209aa990a6237d283a445cc76b535c9b7764
foundation_30series	a7b0542cbbe82b982bfa88b3365fc6382026a9140e7bbfad15d90f8104818a12
llm_architectures_ollama	3116ecb7b5859832f91fce050c01fb767121004700d20f1b9824bcb14fc20980
llm_prose_reference_30series	1cacb94fbc971705a142d945077b26be9b8c4780c446bcc7ab5b662fd3bbc0c7
llm_prose_study_ollama	0256bd45b8ec754c2fbf162f2aeaa0858fa3548cc8f5dbe0f0c77298529a6cfe
llm_prose_study_reference	0e57d47bcbcb5950eb9c821f3b6632e9d4e0fb5875acfd36176de88454765d943
llm_template_architectures_30series	cc0fc00f3f27c1ba4337c6449f10ae6780843e91e1f51b41010c3637baf7b14a
parallel_runner_validation	94a0c8ef793b031ed8e4b3ae872ffff6f304ce37c29a7ac9136f4e40197d8e1d
six_hour_agentic_prose	4b70312c0f8aaec3bb29b185e612c356d0e18ddcaf07f3018ae76b738e2a0a78
six_hour_numeric	d56fedc68e3e4a1ebdf4fc8d2db576149ab28dbf5602a9affa49908ab647e91b
six_hour_numeric_30seconds	34d1bc72655b25233b2de4153752c1fe1a474db3949e9666556e22d5363affc2
six_hour_numeric_extension	d400722b69a70caab08615840a5bea23019e3f9784dc243ef5c777897ceb2e88
six_hour_structured_controls	a3656550ba05e697ec0e99a5e084e93d0bce464b0d101f693d14f5302fb8634c

Refresh and completion criteria

Refresh this report after workers complete or readiness changes using `.venv/bin/python results/report/generate_report.py --scope six-hour` from the repository root. The generator launches no experiments. Completed configured execution requires actual run artifacts and separately reviewed audit/error outcomes. A blocked or partial snapshot remains labeled as such; the six-hour subset does not complete unimplemented dissertation requirements.